

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1. (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Simplify $\frac{1+a^{-1}}{1+a^{-3}}$

2

(b) If $y = \sec x$

i) show that $\frac{dy}{dx} = \sec x \tan x$

2

ii) find $\frac{d^2y}{dx^2}$ in terms of $\sec x$

2

(c) Let $A(-3, 6)$ and $B(1, 10)$ be points on a number plane.

2

Find the coordinates of point C , which divides the interval AB externally in the ratio 5:3

(d) Evaluate $\int_{-3}^3 \frac{1}{9+x^2} dx$

2

(e) If $\frac{dy}{dx} = 1 + y$ and when $x = 0$ $y = 2$, find an expression for y in terms of x

2

Question 2. (12 marks) Use a SEPARATE writing booklet.

Marks

(a) i) Express $\sqrt{2} \sin x + \sqrt{2} \cos x$ in the form $R \sin(x + \alpha)$,
where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$.

2

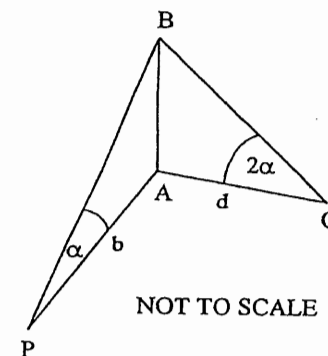
ii) Hence sketch $y = \sqrt{2} \sin x + \sqrt{2} \cos x$ for $0 \leq x \leq 2\pi$
(Show intercepts and endpoints clearly).

2

iii) Hence find the value(s) of k for which $\sqrt{2} \sin x + \sqrt{2} \cos x = k$ has 3 solutions
in the domain of $0 \leq x \leq 2\pi$.

1

(b) From a point P , a distance of b metres south of a tower AB , the angle of elevation to the top of the tower B is α . From point Q , a distance of d metres due east of the tower the angle of elevation to the top of the tower is 2α .



i) Show that $b \tan \alpha = d \tan 2\alpha$

2

ii) Find the height of the tower in terms of d and b

3

iii) If the distance PQ is $d\sqrt{10}$ metres find α

2

Question 3. (12 marks) Use a SEPARATE writing booklet.

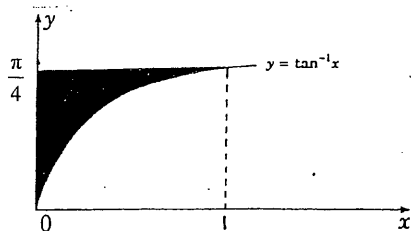
Marks

(a) Solve the inequation $\frac{x}{x-3} < 4$

3

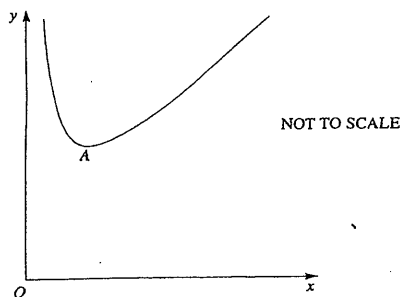
(b) Show that the shaded area bounded by the curve $y = \tan^{-1} x$ the y -axis and the line $y = \frac{\pi}{4}$ is given by $A = \frac{1}{2} \log_e 2$ units².

3



(c) Consider the function $f(x) = 4x + \frac{1}{x}$ for $x > 0$.

The diagram below shows the graph of the function and the stationary point $A\left(\frac{1}{2}, 4\right)$



i) What is the largest domain for which $f(x)$ has an inverse function $f^{-1}(x)$?

1

ii) Copy or trace the graph of $y = f(x)$ into your Writing Booklet. On the same set of axes, draw the graph of $y = f^{-1}(x)$.

2

iii) Find the inverse function $f^{-1}(x)$.

3

Question 4. (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Consider the equation $2x^3 + x^2 - 15x - 18 = 0$.

3

One of the roots of this equation is positive and equals the product of the other two roots.

Find the roots of this equation.

(b) When the polynomial $P(x)$ is divided by $1 - x^2$ it gives $4 - x$ as the remainder.

2

What is the remainder of $P(x)$ when divided by $1 + x$?

(c) The equation $\sin x = x^2 - 10$ has a root close to $x = \pi$.

2

Use one application of Newton's Method to give a better approximation.

(correct your answer to 4 decimal places).

(d) Prove by induction that $\cos(x + n\pi) = (-1)^n \cos x$ for integer $n \geq 1$.

5

16

- (a) AB and CD are two intersecting chords of a circle and CD is parallel to the tangent to the circle at B.
- (i) Draw a neat sketch of the above information. 1
- (ii) Prove that AB bisects $\angle CAD$. 3
- (b) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.
- (i) Derive the equation of the tangent to the parabola at the point P . 2
- (ii) Find the coordinates of the point of intersection T of the tangents to the parabola at P and Q . 2
- (iii) You are given that the tangents at P and Q intersect at 45° . 2
- Show that $p - q = 1 + pq$, where $p > q$.
- (iv) Find the locus of T when the tangents at P and Q intersect as given in (iii). 2

- (a) The rate at which a body cools in air is given by the difference between the air temperature, T° , at any time t minutes and the temperature, A° , of the surrounding air.
- This rate is given by the differential equation:
- $$\frac{dT}{dt} = k(T - A) \quad \text{where } k \text{ is constant.}$$
- i) Show by differentiation that $T = A + Pe^{kt}$, where P is constant, is a solution of the differential equation. 2
- ii) A hot cup of coffee cools from 90°C to 70°C in 8 minutes, the temperature of the air being 22°C . 2
- Find the time required for the cup of coffee to cool to a drinkable temperature of 60°C .
- iii) Sketch the graph of T as a function of t and describe the behaviour of T as t becomes large. 2
- (b) A particle moves in a straight line in simple harmonic motion.
- The acceleration in metres per second per second is given by $\ddot{x} = 2 - 3x$, where x metres is the displacement of the particle from the origin.
- Initially the particle is at $x = 1$ moving with a velocity of $\sqrt{5}$ m/s.
- i) Using integration show that the velocity v m/s of the particle is given by 2
- $$v^2 = 4 + 4x - 3x^2$$
- ii) Find the amplitude of the motion. 1
- iii) Find the centre of motion. 1
- iv) Find the maximum speed of the particle. 1
- v) Find the period of the motion. 1

$$\begin{aligned} \text{a) } \frac{1+a^{-1}}{1+a^{-3}} &= \frac{1+\frac{1}{a}}{1+\frac{1}{a^3}} \\ &= \frac{a+1}{a} \times \frac{a^3}{a^3+1} \\ &= \frac{a^2}{a^2-a+1} \quad \checkmark \end{aligned}$$

Many did not complete factorisation

$$\begin{aligned} \text{b) i) } y &= \sec x \\ &= (\cos x)^{-1} \\ \frac{dy}{dx} &= -(\cos x)^{-2} \times -\sin x \quad \checkmark \\ &= \frac{\sin x}{\cos^2 x} \\ &= \tan x \sec x \quad \checkmark \end{aligned}$$

Show more than more than repeat value of standard integrals

$$\begin{aligned} \text{ii) } \frac{dy}{dx} &= \tan (\tan x \sec x) + \sec^3 x \quad \checkmark \\ &= \sec x (\tan^2 x + \sec^2 x) \\ &= 2 \sec^3 x - \sec x \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{c) } C &= \left(\frac{5 \times 1 - 3 \times -3}{5-3}, \frac{5 \times 0 - 3 \times 6}{5-3} \right) \quad \checkmark \\ &= (7, 16) \quad \checkmark \end{aligned}$$

many still need to learn formula

$$\begin{aligned} \text{d) } \int_{-3}^3 \frac{1}{9+x^2} dx &= \left[\frac{1}{3} \tan^{-1} \frac{x}{3} \right]_{-3}^3 \quad \checkmark \\ &= \frac{1}{3} \left(\tan^{-1} 1 - \tan^{-1} (-1) \right) \\ &= \frac{1}{3} \left(\frac{\pi}{4} - -\frac{\pi}{4} \right) \\ &= \frac{\pi}{6} \quad \checkmark \end{aligned}$$

$$\text{e) } \frac{dy}{dx} = 1+y$$

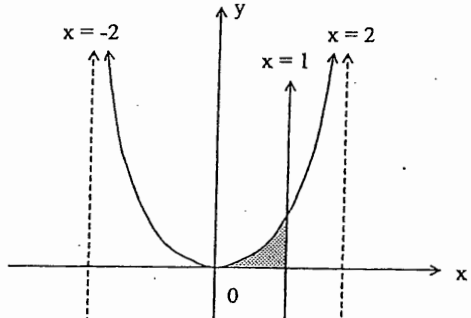
$$\begin{aligned} \frac{dy}{dy} &= \frac{1}{1+y} \\ x &= \ln(1+y) + C \quad \checkmark \\ x=0, y=2 &\therefore C = -\ln 3 \end{aligned}$$

$$\begin{aligned} x &= \ln \left(\frac{1+y}{3} \right) \\ e^x &= \frac{1+y}{3} \\ y &= 3e^x - 1 \quad \checkmark \end{aligned}$$

Question 7. (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) The shaded area in the diagram below represents the area bounded by the curve $y = \frac{x^2}{\sqrt{4-x^2}}$, the x-axis and the line $x = 1$.



Using the substitution $x = 2 \sin \theta$, find the shaded area.

4

- (b) Use the substitution $u = \tan 2x$ to show that $\int_0^{\frac{\pi}{8}} \frac{2 \sec^2 2x}{\sqrt{2 - \tan^2 2x}} dx = \frac{\pi}{4}$

3

- (c) A projectile is fired horizontally with speed $v \text{ ms}^{-1}$ from a point h metres above horizontal ground.

- i) Prove that it will reach the ground after $\sqrt{\frac{2h}{g}}$ seconds.

2

- ii) If it does so at an angle of 60° to the horizontal, show that $3v^2 = 2gh$

3

Hint $\frac{dy}{dx} = \tan 120^\circ$

Question 1.

$$\begin{aligned} 2) \frac{1+a^{-1}}{1+a^{-3}} &= \frac{1+\frac{1}{a}}{1+\frac{1}{a^3}} \\ &= \frac{a+1}{a} \times \frac{a^3}{a^3+1} \quad \checkmark \\ &= \frac{a^2}{a^2+a+1} \quad \checkmark \end{aligned}$$

Many did not
cancel
factorisation

$$\begin{aligned} 3) i) y &= \sec x \\ &= (\cos x)^{-1} \\ \frac{dy}{dx} &= -(\cos x)^{-2} \times -\sin x \quad \checkmark \\ &= \frac{\sin x}{\cos^2 x} \\ &= \tan x \sec x \quad \checkmark \end{aligned}$$

Show more
more than
repeat table
of standard
integrals

$$\begin{aligned} ii) \frac{d^2 y}{dx^2} &= \tan(\tan x \sec x) + \sec^3 x \quad \checkmark \\ &= \sec x (\tan^2 x + \sec^2 x) \\ &= 2\sec^3 x - \sec x \quad \checkmark \end{aligned}$$

$$\begin{aligned} 4) C &= \left(\frac{5 \times 1 - 3 \times 3}{5 - 3}, \frac{5 \times 10 - 3 \times 6}{5 - 3} \right) \quad \checkmark \\ &= (7, 16) \quad \checkmark \end{aligned}$$

many still
need to learn
formula

$$\begin{aligned} 5) \int_{-3}^3 \frac{1}{9+x^2} dx &= \frac{1}{3} \tan^{-1} \frac{x}{3} \Big|_{-3}^3 \quad \checkmark \\ &= \frac{1}{3} (\tan^{-1} 1 - \tan^{-1} -1) \\ &= \frac{1}{3} \left(\frac{\pi}{4} - -\frac{\pi}{4} \right) \\ &= \frac{\pi}{6} \quad \checkmark \end{aligned}$$

Question 1.

$$e) \frac{dy}{dx} = 1+y$$

$$\frac{dx}{dy} = \frac{1}{1+y}$$

$$x = \ln(1+y) + C \quad \checkmark$$

$$x=0, y=2 \quad C = -\ln 3$$

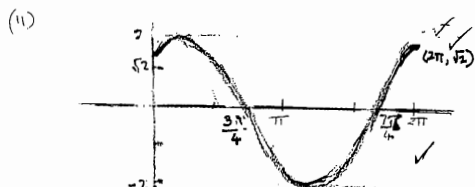
$$x = \ln\left(\frac{1+y}{3}\right)$$

$$e^x = \frac{1+y}{3} \quad \checkmark$$

$$y = 3e^x - 1$$

Question 2.

a) (i) $\sqrt{5}\sin x + \sqrt{2}\cos x = 2\sin(x + \frac{\pi}{4})$ ✓



(iii) $k = \frac{-2}{\sqrt{2}}$

b) (i) $\tan d = \frac{AB}{b}$, $\tan 2d = \frac{AB}{d}$ ✓

$b \tan d = d \tan 2d$ ✓
(ii) let $AB = h$
 $h = d \tan 2d$
 $= d \left(\frac{2 \tan d}{1 - \tan^2 d} \right)$ ✓

$= d \left(\frac{2 \frac{h}{b}}{1 - \frac{h^2}{b^2}} \right) = \frac{2 h d b}{b^2 - h^2}$ ✓

$b^2 - h^2 = 2 d b$

$h^2 = b^2 - 2 d b$

$h = \sqrt{b^2 - 2 d b}$ ✓

(iii) $b^2 + d^2 = 10 d^2$ Pythagoras!
 $\tan \alpha = \frac{h}{b} = \frac{\sqrt{b^2 - 2 d b}}{b} = \frac{\sqrt{b^2 - 2 b \cdot \frac{2}{\sqrt{3}}}}{b}$ ✓
 $= \frac{1}{\sqrt{3}}$
 $\alpha = \frac{\pi}{6}$ ✓

well done by majority

more care required in terms of intercept and endpoints for many students

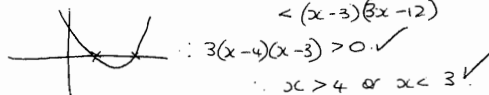
generally well executed

most attempted the early part of question, but some students could not achieve the desired result.

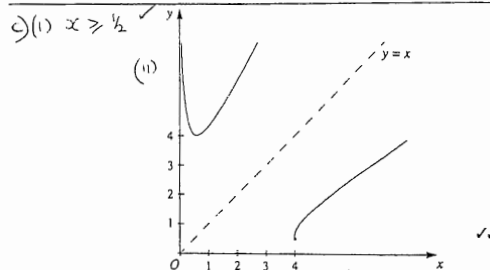
generally performed well by students who obtained the result in part (ii)

Question 3

a) $\frac{x}{x-3} < 4$ $x(x-3) < 4(x-3)^2$ ✓
 $0 < (x-3)[4(x-3)-x]$
 $< (x-3)(8x-12)$



b) $A = \int_0^{\pi/4} \tan y dy$ ✓
 $= -\ln(\cos y) \Big|_0^{\pi/4}$ ✓
 $= -\ln \frac{1}{\sqrt{2}}$ ✓
 $= \frac{1}{2} \ln 2$ ✓



(iii) $y = 4x + \frac{1}{x}$ for $x \geq \frac{1}{2}$
Inverse: $x = 4y + \frac{1}{y}$ for $y \geq \frac{1}{2}$ ✓
 $xy = 4y^2 + 1$
 $4y^2 - xy + 1 = 0$

$y = \frac{x \pm \sqrt{x^2 - 16}}{8}$
 $y = \frac{x + \sqrt{x^2 - 16}}{8}$ or $\frac{x - \sqrt{x^2 - 16}}{8}$ ✓

For the appropriate range, $y \geq \frac{1}{2}$

Many students expand and fail to realise they can factorise $(x-3)$ from both sides.

largest domain $x > 1$

Many sketches an inverse which not a function

Question 4

Let roots be $\alpha, \beta, \alpha\beta$

$\alpha + \beta + \alpha\beta = -1/2$ ① ✓
 $\alpha\beta + \alpha^2\beta + \alpha\beta^2 = -1/2$ ② ✓

$\alpha^2\beta^2 = 9$

$\alpha\beta = \pm 3$ but from question $\alpha\beta = 3$ ✓

From ① $\alpha + \beta = -3/2$ } solve simultaneously
 $\alpha\beta = 3$

∴ Roots are $-\frac{3}{2}, -2$ and 3 . ✓

b) $P(x) = (1-x^2)Q(x) + (4-x)$
 $= (1-x)(1+x)Q(x) + (4-x)$ ✓

$P(-1) = 4 - (-1) = 5$

∴ remainder is 5. ✓

c) $P(x) = \sin x - x^2 + 10$
 $P'(x) = \cos x - 2x$ ✓
 $x_2 = \pi - \left(\frac{10 - \pi^2}{-1 - 2\pi} \right)$ ✓
 $= 3.1594$

d) If $n=1$ LHS = $\cos(x+\pi)$
 $= \cos x \cos \pi - \sin x \sin \pi$
 $= -\cos x$
RHS = $-\cos x$
True for $n=1$. ✓

Assume true for $n=k$

$\cos(x+k\pi) = (-1)^k \cos x$ ✓

Consider $n=k+1$

$\cos(x+(k+1)\pi) = \cos(x+k\pi + \pi)$ ✓
 $= \cos(x+k\pi)\cos\pi - \sin(x+k\pi)\sin\pi$
 $= (-1)^k \cos x \cdot (-1) - 0$
 $= (-1)^{k+1} \cos x$ ✓

which is of the same form as for $n=k$

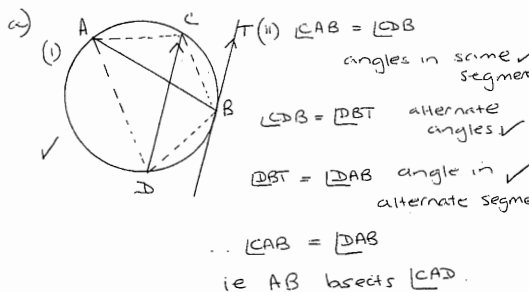
If true for $n=k$, also true for $n=k+1$. Since true for $n=1$, true for $n=2$ and hence all following positive integers. ✓

with trig. "think" expansions

concluding statement was poorly written by many students.

Many students did not show LHS = RHS for $n=1$

Question 5



b) (i) $y = \frac{x^2}{4a}$
 $\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a} = p$ at $(2ap, ap^2)$
 Eqn is $y - ap^2 = p(x - 2ap)$
 $y = px - ap^2$ ✓

(ii) $px - ap^2 = qx - aq^2$
 $(p-q)x = ap^2 - aq^2$
 $= a(p-q)(p+q)$
 $x = a(p+q)$ ✓
 $y = pa(p+q) - ap^2$
 $= apq$ ✓

$T = (a(p+q), apq)$ ✓

(iii) $\tan 45^\circ = \frac{p-q}{1+pq}$ ✓
 $1 = \frac{p-q}{1+pq}$ ✓

Diagrams need to be neat, clear and $\frac{1}{2}$ page. Use a ruler to draw straight lines. Make sure all information in question is placed on diagram.
 * If you assumed chords passed through centre - zero marks for proof.

Parts (i), (ii), (iii) well done.
 must state $\tan 45^\circ = 1$. We do not assume you know this - especially with 'SHOW' questions.

(iv) $(p-q)^2 = (1+pq)^2$
 $= 1 + pq^2 + 2pq$ ✓
 $(p+q)^2 - 4pq = 1 + pq^2 + 2pq$
 $\frac{x^2}{a^2} - 4\frac{y}{a} = 1 + \frac{y^2}{a^2} + 2\frac{y}{a}$ ✓
 $x^2 - 4ay = (a+y)^2$ ✓

$x = a(p+q)$ $y = apq$

$\frac{x}{a} = (p+q)$

$\frac{x^2}{a^2} = (p^2 + q^2 + 2pq)$
 $= (p-q)^2 + 2pq + 2pq$

$= (p-q)^2 + 4pq$

$= (1+pq)^2 + 4pq$

$= (1 + \frac{y}{a})^2 + \frac{4y}{a}$

$\frac{x^2}{a^2} = 1 + \frac{2y}{a} + \frac{y^2}{a^2} + \frac{4y}{a}$

$x^2 = a^2 + 2ay + y^2 + 4ay$

$x^2 = a^2 + y^2 + 6ay$

use $1+pq = p-q$

use $\frac{y}{a} = pq$

To do better with locus problems, you need to practise a variety of problem involving different techniques.
 An alternative way to do (iv)

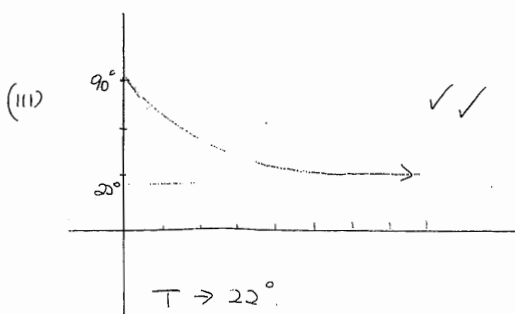
Question 6

Ans
 (i) $T = A + Pe^{kt}$
 $\frac{dT}{dt} = kPe^{kt}$ ✓
 $= k(T-A)$ ✓

(ii) $t=0, T=90^\circ$ $90 = 22 + P$
 $P = 68$ ✓
 $t=8, T=70^\circ$ $70 = 22 + 68e^{8k}$
 $e^{8k} = \frac{48}{68}$
 $= 0.70588336$
 $k = -0.04$ ✓

$60 = 22 + 68e^{kt}$ ✓

$t = 13 \text{ mins}$



generally well done

some did not interpret initial conditions correctly.

$k = \frac{\ln(\frac{48}{68})}{8}$ also gave the mark.

ignored seconds as units in some answers

many did not show 90° y-intercept, nor the asymptote as $t \rightarrow \infty$

also some strange interpretations of what an exponential graph looks like!

Question 6(b)

$\ddot{x} = 2-3x$ $x=1, v=\sqrt{3}$

(i) $\frac{d}{dx}(\frac{1}{2}v^2) = 2-3x$

$\frac{1}{2}v^2 = 2x - \frac{3}{2}x^2 + c$ ✓

When $x=1, v=\sqrt{3}$

$\frac{1}{2} \times 3 = 2 - \frac{3}{2} + c$

$c = 2$ ✓

$\therefore \frac{1}{2}v^2 = 2x - \frac{3}{2}x^2 + 2$

$v^2 = 4x - 3x^2 + 4$

$v^2 = 4 + 4x - 3x^2$ ✓

(ii) When $v=0$, $4 + 4x - 3x^2 = 0$

$3x^2 - 4x - 4 = 0$

$(3x+2)(x-2) = 0$

$x = -\frac{2}{3}, 2$

Amplitude = $\frac{1}{2} \times 2\frac{2}{3}$

$= 1\frac{1}{3} \text{ m}$ ✓

(iii) $a = -3(x - \frac{2}{3})$

Centre of motion is $x = \frac{2}{3}$ ✓

(iv) Maximum speed occurs at centre of motion.

When $x = \frac{2}{3}$, $v^2 = 4 + 4 \times \frac{2}{3} - 3 \times \frac{4}{9}$

$= 5\frac{1}{3}$

$v = \pm \frac{4}{\sqrt{3}}$

Max. speed is $\frac{4}{\sqrt{3}}$ or $\frac{4\sqrt{3}}{3} \text{ m/s}$ ✓ 2.3094

(v) $n^2 = 3$, $n = \sqrt{3}$

$\therefore \text{Period} = \frac{2\pi}{\sqrt{3}}$ ✓

generally well done

many found extremes but did not halve the distance between these extremes correctly!

for (ii) and (iii) many tried (poorly) to complete the square to put into form $v^2 = n^2(a^2 - x^2)$

generally well done.
 leave as a surd unless a number of d.p.s is asked for.

many had trouble deducing $n^2 = 3$.

Question 7

$$a) \text{ Area} = \int_0^1 \frac{x^2 dx}{\sqrt{4-x^2}} \quad x = 2\sin\theta \quad \frac{dx}{d\theta} = 2\cos\theta$$

students did not replace dx with dθ

$$= \int_0^{\pi/6} \frac{4\sin^2\theta \cdot 2\cos\theta d\theta}{\sqrt{4-4\sin^2\theta}}$$

students did not cancel 2cosθ

$$= \int_0^{\pi/6} \frac{4\sin^2\theta \cdot 2\cos\theta d\theta}{2\cos\theta}$$

$$= 2 \int_0^{\pi/6} 1 - \cos 2\theta d\theta$$

$$= 2 \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/6}$$

$$= 2 \left[\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right]$$

$$= \frac{2\pi - 3\sqrt{3}}{6}$$

$$b) \int_0^{\pi/6} \frac{2\sec^2 2x dx}{\sqrt{5-\tan^2 2x}} \quad u = \tan 2x \quad \frac{du}{dx} = 2\sec^2 2x$$

well done

$$= \int_0^1 \frac{du}{\sqrt{2-u^2}}$$

$$= \left[\sin^{-1} \frac{u}{\sqrt{2}} \right]_0^1$$

$$= \frac{\pi}{4}$$

students did not change integration boundaries to 1 & 0.

Question 7

$$i) \dot{x} = 0 \quad \ddot{y} = -g$$

$$\dot{x} = V \cos \theta$$

$$= V \quad (\text{speed} = 0)$$

$$\dot{y} = -gt$$

$$y = h - \frac{g}{2} t^2$$

$$y = h - \frac{g}{2} t^2$$

well done!

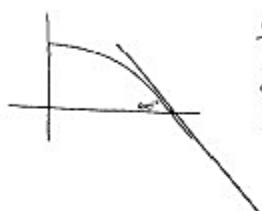
It reaches ground when $y = 0$.

$$0 = h - \frac{g}{2} t^2$$

$$t^2 = \frac{2h}{g}$$

$$t = \sqrt{\frac{2h}{g}}$$

(ii)



$$\frac{dy}{dx} = \tan 120^\circ = -\sqrt{3}$$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$$

$$= -gt \cdot \frac{1}{v}$$

$$= -\frac{gt}{v}$$

$$= -\frac{g}{\sqrt{g}} \cdot \sqrt{\frac{2h}{g}}$$

$$-\sqrt{3} = -\frac{g}{\sqrt{g}} \cdot \sqrt{\frac{2h}{g}}$$

$$3v^2 = g^2 \frac{2h}{g}$$

$$3v^2 = 2gh$$